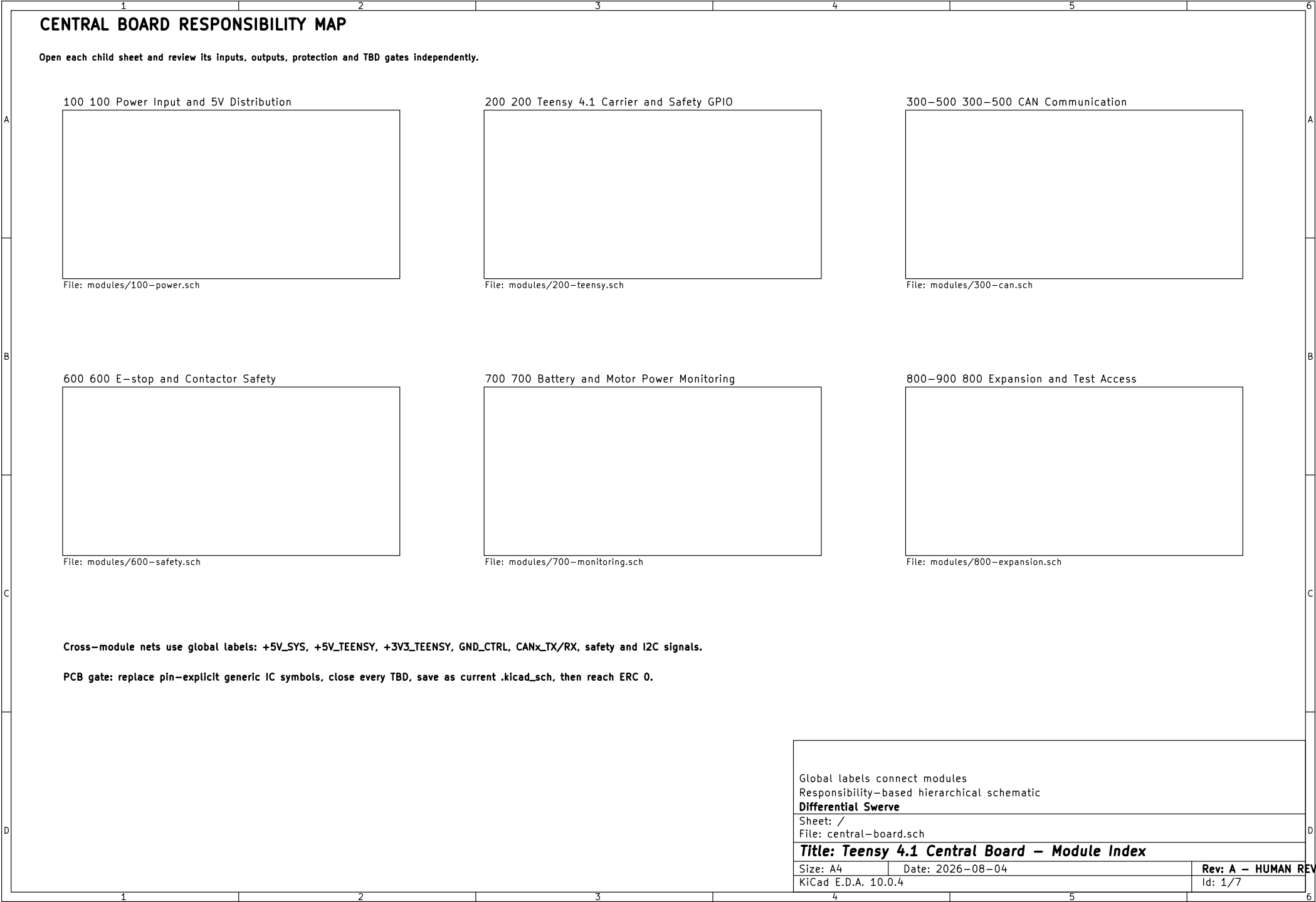
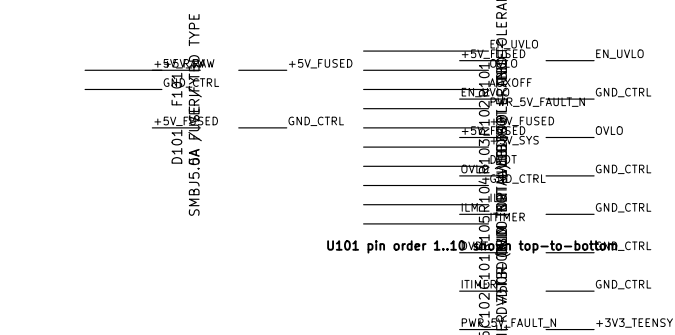
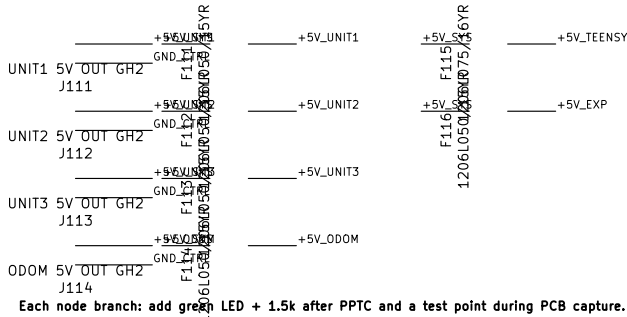




100 POWER INPUT / eFUSE / STAR DISTRIBUTION



CAUTION: OVLO must include rail + comparator + resistor tolerance; nominal 5.45V is too close to Teensy VIN max 5.5V.



Each node branch: add green LED + 1.5k after PPTC and a test point during PCB capture.

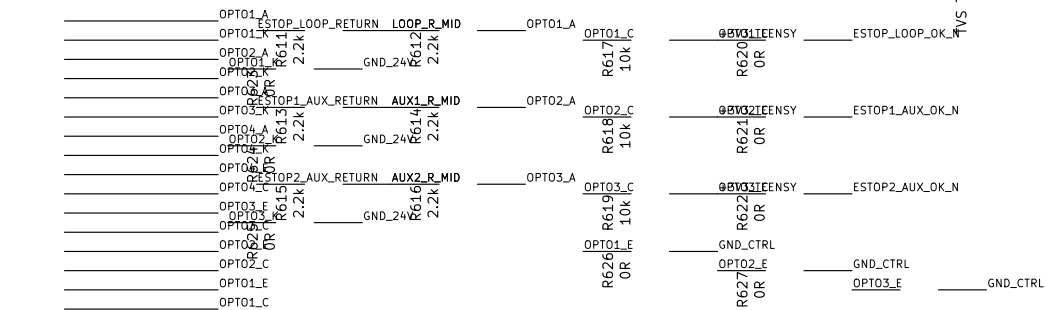
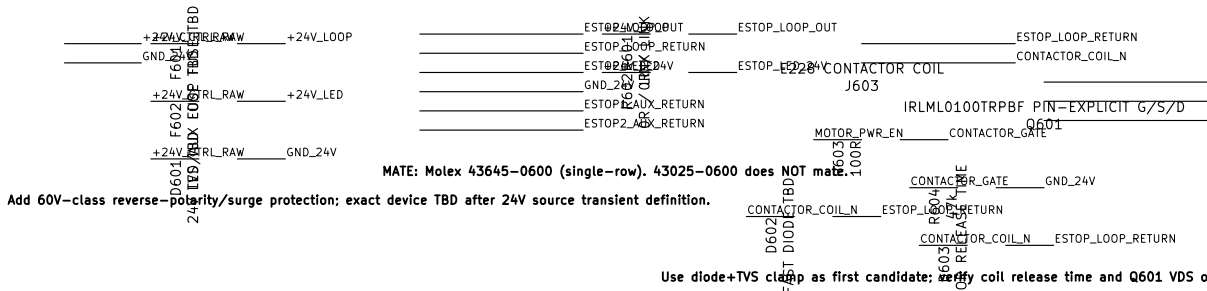
Global labels connect modules
Responsibility-based hierarchical schematic

Differential Swerve
Sheet: /100 100 Power Input and 5V Distribution/
File: 100-power.sch

Title: 100 Power Input and 5V Distribution

Size: A4	Date: 2026-08-04	Rev: A – HUMAN REVIEW DRAFT
KiCad E.D.A. 10.0.4		Id: 2/7

600 E-STOP / CONTACTOR (HARDWARE PATH + ISOLATED MONITOR)



U601 pin order follows LTV-847S datasheet; CH4 spare. Each external 24V input uses two series resistors.

2.2k+2.2k gives about 5mA at 24V, matching the LTV-847S minimum-CTR test condition. Check resistor dissipation and input range.



Ground-domain assumption: GND_24V and GND_CTRL join at protected control-power entry. Verify before layout.

Global labels connect modules
Responsibility-based hierarchical schematic

Differential Swerve
Sheet: /600 600 E-stop and Contactor Safety/
File: 600-safety.sch

Title: 600 E-stop and Contactor Safety

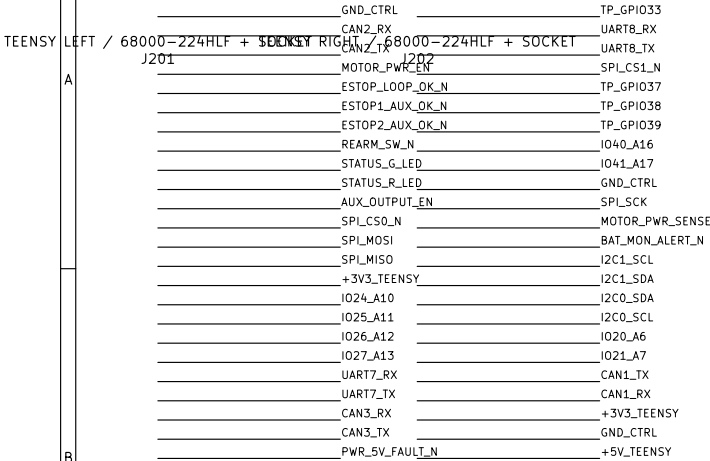
Size: A4
KiCad E.D.A. 10.0.4

Date: 2026-08-04

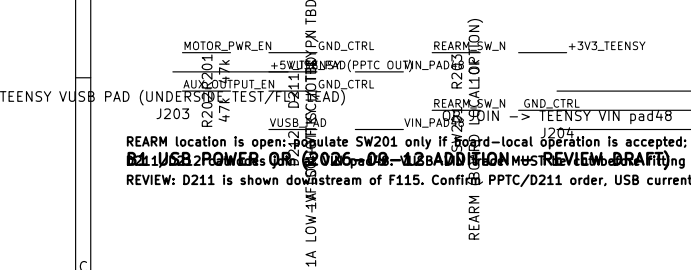
Rev: A - HUMAN REVIEW DRAFT

Id: 3/7

200 TEENSY 4.1 SOCKETS (TOP-VIEW PIN MAP)



J201 socket pads 1..24 = Teensy GND,0..12,3V3,24..32.
J202 socket pads 25..48 = Teensy 33..41,GND,13..23,3V3,GND,VIN.
MECHANICAL: two 1x24 sockets, 17.78mm row spacing; keep entire underside free of parts/exposed pads.
ASSEMBLY: cut Teensy V05B-VIN link before simultaneous external 5V and USB connection.



Global labels connect modules
Responsibility-based hierarchical schematic

Differential Swerve
Sheet: /200 200 Teensy 4.1 Carrier and Safety GPIO/
File: 200-teensy.sch

Title: 200 Teensy 4.1 Carrier and Safety GPIO

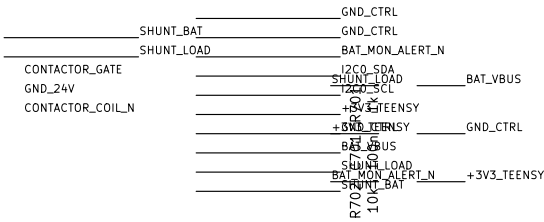
Size: A4 Date: 2026-08-04

KiCad E.D.A. 10.0.4

Rev: A - HUMAN REVIEW DRAFT

Id: 4/7

700 BATTERY MONITOR (EXTERNAL KELVIN SHUNT)



MANDATORY: fusible resistors/small fuses at the battery-side origins of BOTH Kelvin sense wires; board-side resistors cannot protect the harness.
100A/75mV shunt dissipates 5.17W at 83A. Confirm continuous/peak current and thermal mounting before purchase.

Global labels connect modules
Responsibility-based hierarchical schematic

Differential Swerve

Sheet: /700 700 Battery and Motor Power Monitoring/
File: 700-monitoring.sch

Title: 700 Battery and Motor Power Monitoring

Size: A4

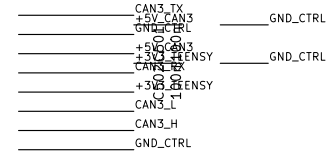
Date: 2026-08-04

Rev: A - HUMAN REVIEW DRAFT

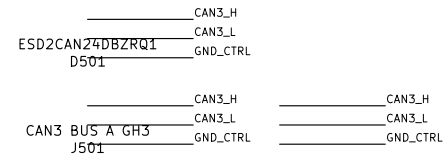
KiCad E.D.A. 10.0.4

Id: 5/7

500 CAN3 – DRIVE CAN-FD 1M/2M



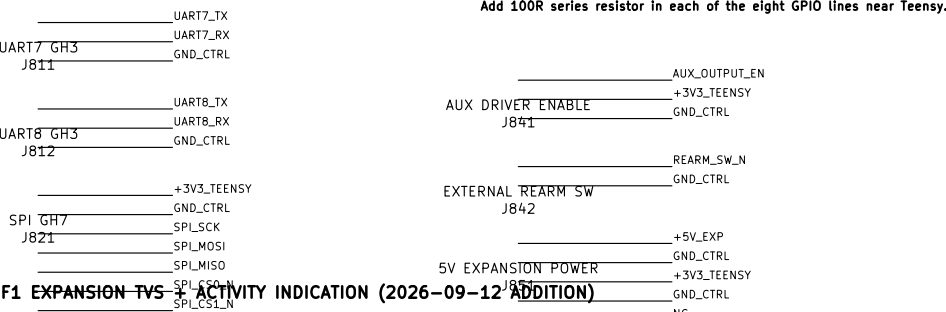
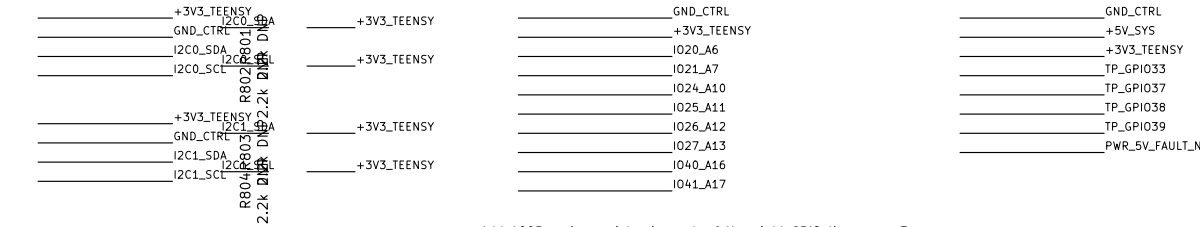
U pin order: TXD,GND,VCC,RXD,VIO,CANL,CANH,S; S tied low.



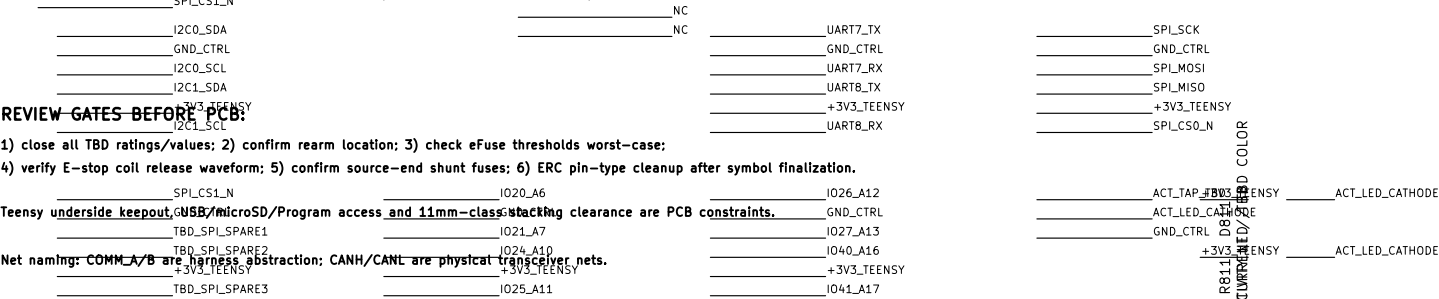
VIN_PAD48
NC
NC

Id: 6/7

800 EXPANSION / 900 TEST ACCESS



F1 EXPANSION TVS - ACTIVITY INDICATION (2026-09-12 ADDITION)



REVIEW GATES BEFORE PCB:

- 1) close all TBD ratings/values; 2) confirm rearm location; 3) check eFuse thresholds worst-case;
- 4) verify E-stop coil release waveform; 5) confirm source-end shunt fuses; 6) ERC pin-type cleanup after symbol finalization.

Teensy underside keepout, USB/microSD/Program access and 11mm-class stacking clearance are PCB constraints.

Net naming: COMM_A/B are harness abstraction; CANH/CANL are physical transceiver nets.

THIS IS A REVIEW DRAFT. Generic pin-explicit symbols are intentional where final custom library symbols are not yet frozen.

REVIEW: SRV05-4 vendor/ratings and exact channel assignment are TBD. Six parts listed by requirements; shown pin-explicit. Activity LEDs: direct signal taps via digital transistor, no new Teensy GPIO. Transistor PN, polarity, and number of taps TBD.

REVIEW: CAN/UART idle polarity may cause constant illumination; choose observed signal/tap topology after bench validation.

Global labels connect modules
Responsibility-based hierarchical schematic

Differential Swerve

Sheet: /800-900 800 Expansion and Test Access/
File: 800-expansion.sch

Title: 800 Expansion and Test Access

Size: A4
Date: 2026-08-04

KiCad E.D.A. 10.0.4

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